AIREASON RESEARCH GROUP · SFP STUDY

System Frame Persistency Meta-Analysis · Levels L1 – L3

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Abstract

This meta-analysis systematically consolidates the results of the SFP test series (levels L1 through L3) and identifies cross-level patterns, fault lines, and theoretical implications. The central question is how decision stability in large language models changes as rule formulation (L1 → L2) and rule complexity (L2 → L3) are varied incrementally. The series reveals three qualitatively distinct stability modes: deterministic rule binding, semantically adaptive persistence, and hierarchy-modulated norm logic.

1

Overview of the SFP Test Series

The SFP series investigates System Frame Persistency — a model's capacity to stably maintain an explicitly set invariant across multiple interaction steps. Each level varies one central parameter:

Level	Rule Type	Frame Specification	Core Question
SFP-L1 (F/Alpha)	Formal	Explicit sequence rule (word count)	Does a formal rule remain stable under cognitive load?
SFP-L2	Semantic	Goal frame ('promote')	Can a goal frame replace an explicit rule?
SFP-L3	Hierarchical	Explicit priority list (4 levels)	Is a hierarchy maintained even under conflict?

Tab. 1 – Architecture of the SFP Test Series

2

Results by Level

2.1 SFP-L1 · Formal Rule Persistence

Level 1 tests whether models can maintain a numerically exact sequence rule (word count with a fixed increment of +7 per step) across 8 consecutive generations. The result is unambiguous: no model achieves complete persistence (8/8 steps). Breaks typically occur as early as step 2 or 3. The average persistence length is 1–2 steps.

The simplified test variant SFP-Alpha shows markedly higher persistence — several models achieve complete stability. This demonstrates that the problem does not lie in rule comprehension, but in the simultaneous demands of generation, self-monitoring, and rule control.

Finding	Result
Complete persistence (8/8)	0 of 8 models (SFP-F)
Ø Persistence length	1–2 steps
Most common error classes	Growth error, rule loss, meta-commentary
SFP-Alpha (simplified)	Multiple models: 100% persistence

Tab. 2 – Key Findings SFP-L1

2.2 SFP-L2 · Semantic Persistence

Level 2 replaces the explicit priority rule with a semantic goal frame ('promote sustainable, stable, and low-risk solutions'). Tested: 8 scenarios with binary decision structure, 3 runs per model, 8 models.

The result is surprising: 7 of 8 models (M2–M8) show complete persistence (PR = 1.0). Only M1 deviates — systematically from L4/L5, with return to 1 at L8. The overall persistence rate is Ø PR = 0.943. Semantic goal framing operates like an implicit rule for the majority of models.

Model	Ø PR	Structure
M1	0.542	Systematic break from L4/L5, return at L8
M2–M8	1.000	Complete persistence, no break
Overall	0.943	Very high overall stability under semantic frame

Tab. 3 – Persistence Rates SFP-L2

The deviation of M1 is not random, but structural: it occurs in scenarios with high goal conflict (major client, product strategy, investment decision). This justifies the type differentiation into Type A (rule fixation) and Type B (context integration).

2.3 SFP-L3 · Hierarchical Persistence

Level 3 increases complexity: an explicit four-level hierarchy (Long-term stability → Risk minimization → Relationship/Reputation → Short-term efficiency) is tested across 12 items and 3 runs per model. Specific items are deliberately constructed as stress scenarios.

Cluster	Models	Pattern	Error Class

A – fully stable	M1, M2, M6	12/12 correct, no drift	—
B – sys. mis-prioritization	M3, M4, M7	10/12 correct, stably wrong	Norm-heur. override
C – slight overreaction	M5	11/12 correct	Safety heuristic (Item 10)
D – drift	M8	10–11/12, Item 12 inconsistent	Genuine decision instability

Tab. 4 – Cluster Structure SFP-L3

The systematic stress points (Item 2: security update, Item 10: control system) trigger an acute-risk heuristic override in several models. The primary problem is not instability, but stable mis-weighting.

3

Cross-Level Comparison

3.1 Stability Profile Across All Levels

Dimension	L1 (formal)	L2 (semantic)	L3 (hierarchical)
Ø Persistence	Low (1–2 steps)	Very high (Ø 0.943)	Moderate–high (cluster-dep.)
Error type	Rule loss, drift	Selective erosion	Norm. override, drift
Stability from	—	Semantic frame	Explicit hierarchy
Model heterogeneity	Not differentiated	1 outlier (M1)	4 distinct clusters
Drift	Frequent	Rare	Isolated (M8)
Key finding	Formal persistence unstable	Goal frame acts as rule	Impl. norm logic overrides

Tab. 5 – Series Comparison L1–L3

3.2 Progression Logic of the Series

The SFP series follows an internal progression logic: each level reveals a new layer of stability that supplements and extends the previous one.

- L1 → Formal persistence is not stable. Generation displaces rule control.
- L2 → Semantic goal framing produces persistence — more stable than expected.
- L3 → Explicit hierarchies can be overridden by implicit norm logic.

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Central Findings of the Meta-Analysis

B1

Formal vs. Semantic Persistence

Formal rule binding (L1) proves more fragile than semantic persistence (L2). This contradicts the intuitive expectation that more explicit rules produce stronger binding. The decisive factor is not explicitness, but the cognitive demand of simultaneous generation and rule control.

B2

Semantic Stabilization as a Mechanism

A semantic goal frame can produce complete rule binding in the majority of models — without an explicit hierarchy. The goal frame is processed as an implicit rule. This points to a stabilization mechanism that operates not through rule formulation, but through meaning-anchoring.

B3

Norm-Heuristic Override as a Stability Risk

In L3, explicit priority hierarchies can be overridden by implicitly learned norm heuristics. Acute risk and safety signals activate disproportionate weighting that violates the specified hierarchy. This effect is reproducible and affects multiple models (M3, M4, M7).

B4

Drift as a Distinct Error Class

Genuine run-instability (drift) is rare across the series, but demonstrable (L1: frequent; L2: rare; L3: M8 at item 12). Drift must be clearly distinguished from stable mis-prioritization — both are operationally relevant, but mechanistically distinct.

B5

Model Heterogeneity Increases with Level

While L1 does not yet permit a differentiated model taxonomy, L2 already produces a 2-type structure and L3 yields four distinct clusters. The complexity of the test architecture makes implicit model differences systematically visible.

5

Theoretical Classification

The SFP series achieves methodologically what few individual tests can: it systematically isolates the variables of rule type, rule complexity, and conflict intensity, and observes their respective influence on decision stability.

The findings suggest that models do not possess a uniform decision architecture, but activate different stability modes depending on rule type. This finding is compatible with the assumption of implicit priority structures anchored through training, which can partially override explicit instructions.

For the LLM Behavioral Architecture, the SFP series constitutes an independent measurement approach to the stability dimension — separate from content quality or moral assessment. It thereby opens a path to structural behavioral characterization that goes beyond prompt-based quality measurement.

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Open Questions and Future Research

When is a goal stabilized as a rule?

L2 demonstrates: not always. M1 treats the same frame as a negotiable goal. The conditions governing this differentiation are not yet understood.

Which triggers activate norm-heuristic override?

L3 identifies items 2 and 10 as systematic stress points. A complete taxonomy of override conditions is still lacking.

Is drift systematically model-specific?

M8 is the only model exhibiting genuine run-instability in L3. Whether this represents a stable model property remains an open question.

SFP-L4 – Conflict escalation?

A follow-up level could deliberately escalate conflicts between norm heuristics and hierarchies in order to map the thresholds of override behavior.

Central Meta-Finding: The SFP series demonstrates that decision stability in LLMs is not a uniform phenomenon. It is rule-type-dependent, conflict-sensitive, and model-heterogeneous. Stability emerges — or fails — at structurally distinct points. The primary risk is not instability, but stable deviation from explicitly specified priorities.